

Running log — improvements to the class-AB pad driver

sg13cmos5l_cm_ip_single2diff2single, design_considerations/class_ab_pad_driver/improvements

2 October 2026

Session 2026-10-02, Claude (configured model `claude-opus-5-5`; the serving model may differ). Newest entries at the bottom. Times are Europe/Berlin, taken from the session transcript (corrected 15:01: the first versions of several headings carried guessed times, up to $1\frac{1}{4}$ h too late).

Task as given: area, current and loop gain of `d2s_miller` / `d2s_loadcomp` are unsatisfactory, probably because of the linear passives (rhigh degeneration, MOM Miller caps). Look in Edwards, Minch, Cauwenberghs, Hasler (DoNotLitter/TimEdwards, LogDomain, FloatingGate) and Sarpeshkar 1997 for (1) alternatives to resistor source degeneration in the DDA, (2) compensation without linear BEOL capacitance, (3) cascoding with the output devices either doubling as ESD clamps or as separate devices in the slot, (4) externally-linear / internally-nonlinear circuits whose internal nonlinearity is not the subthreshold exponential. End result: proposed improvements to `class_ab_pad_driver`.

Nothing is written to `~/DoNotLitter`. Papers are cited, not copied (chatlog/ref rule).

14:10 — baseline taken from the existing note and the first CACE run

What the CACE run (`verification/cace/first_run_summary.md`) adds to the design note:

- Gain spread 0.476...0.554 over PVT + rhigh corners, 0.41...0.61 in process MC. Cause: the gain is $(R_2 + 2/g_m)/(R_1 + 2/g_m)$; R tracks R , but g_m does not track R .
- Output offset in mismatch MC $-32...+49$ mV (4 degenerated pairs; offset referred through $1/G_m$ of degenerated pairs is large).
- INL 469 mV worst corner; THD -15 dB at 1.25 V output (pairs run out of $I_{tail}\cdot R$ range).
- Output noise $1.6\text{ }\mu\text{V}/\sqrt{\text{Hz}}$ at 1 kHz (degeneration resistors + low G_m).
- Area: the two MOM Miller caps ($2 \times 31 \times 31\text{ }\mu\text{m}$) dominate the non-frame area (1920 of $\sim 2500\text{ }\mu\text{m}^2$).
- Current: 427 μA , of which 240 μA output-stage quiescent, $\sim 180\text{ }\mu\text{A}$ front end + bias.

14:25 — literature read so far (details in `literature_notes.md`)

- Sarpeshkar 1997 thesis (scanned, OCR'd locally on the linked computer), ch. 3 = the wide-linear-range OTA: well input, diode source degeneration, gate degeneration through the mirror diode, bump linearization ($w = 2$ cancels the cubic term). No resistors.
- Edwards thesis §3.5–3.10, Edwards & Cauwenberghs 2000: ELIN/log-domain; names the square law of strong inversion as the other usable internal nonlinearity; matching = "same surround".
- Minch thesis ch. 7: a source follower is linear whatever the device law (MITE via FG source follower, valid above threshold).
- Hasler thesis ch. 4–5, Sánchez-Sinencio FG slides: capacitive dividers / FG inputs widen linear range, but need linear caps and charge control (tunneling/injection or UV).

14:25 — simulation environment

Cloud container: ngspice-42 (Ubuntu), IHP-Open-PDK dev sparse checkout, OpenVAF-reloaded built from source with LLVM 18 for psp103 / r3_cmc / cap_cmomi. (in progress)

14:28 — simulation environment ready, baseline reproduced

ngspice-42 + IHP-Open-PDK dev models + psp103 / r3_cmc / cap_cmomi compiled with OpenVAF-reloaded (LLVM 18, OSDI minor version patched 4 $\rightarrow$ 3 for ngspice 42, as in the first session). `run_d2s.py` functions reproduce `results.txt` exactly (A, 1 k Ω || 100 pF: gain 0.5005, 0.18 mV, nonlin 5.52 mV, T0 66.5 dB, fc 1.62 MHz, PM 74.8°, THD 10 kHz -51.1 dB).

14:40 — DDA unit characterization (sim/units.spice; test bench now in run_improvements.py units)

Each unit is one split-tail PMOS pair; $x = g_p - g_n$ swept ± 0.8 V at input CM 1.40 / 1.65 / 1.90 V.

unit	degeneration	Gm0 @ CM 1.40 / 1.65 / 1.90	f(0.5)/(Gm0·0.5)	CM sensitivity of Gm
unit_r	rhig 0.5 $\times$ 36.4 μ m (108 k Ω), Lt = 2 $\times$ 10 μ A	15.54 / 15.55 / 15.58 μ S	0.996	0.5 %/V
unit_t	hv PMOS 0.5/3 in triode, gate at vss	12.43 / 15.54 / 18.40 μ S	0.992	77 %/V
unit_w	Sarpeshkar well input, 4/4, gates at vss, 1 $\times$ 20 μ A	11.82 / 11.02 / 10.35 μ S	0.991	-27 %/V

The MOS alternatives are linear enough on their own; their problem is common-mode sensitivity (triode R depends on $V_{SG} = \text{source} - \text{gate}$; well-input κ depends on well-to-gate voltage, as Sarpeshkar's §3.3.1 reports). Why that matters for the next step: see 14:43 (a leftover "15:05" from the guessed timestamps, corrected 16:15).

14:43 — matched-pair DDA (MP-DDA)

Idea: four identical units, A (vinp, vref), B (vref, vinn), C1 = C2 (vref, vfb): $f(\text{vinp} - \text{vref}) + f(\text{vref} - \text{vinn}) = 2 f(\text{vfb} - \text{vref})$. With matched units the solution is $v_{\text{out}} - \text{vref} = (\text{vinp} - \text{vinn})/2$ for *any* odd monotonic f (exact for input CM = vref; for a CM error c the error is second order in c). Gain $\frac{1}{2}$ = unit count ratio; no R1/R2, no gm term. This is static ELIN: the input units compress, the feedback units invert the same law. Caveat found analytically: A and C sit at CM $\text{vref} + x/2$, B at $\text{vref} - x/2$. A unit whose Gm depends on CM (k per volt) gives $v \approx u - k u^2/2 \rightarrow \text{unit}_t \approx 96$ mV, $\text{unit}_w \approx 34$ mV of second-order error at $u = 0.5$ V, $\text{unit}_r \approx 0.6$ mV. So in this topology the poly resistor stays the right degenerator. Its area is $\sim 18 \mu\text{m}^2$ per unit, it never was the area problem.

First simulation, everything else unchanged (fold, mirror, class-AB, frames, MOM caps, bias):

variant	load	gain	nonlin	T0 / fc / PM	THD 10 k / 100 kHz
baseline A	1 k 100 p	0.5005	5.52 mV	66.5 dB / 1.62 MHz / 74.8°	-51.1 / -50.2 dB
MP-DDA, MOM	1 k 100 p	0.4998	1.04 mV	66.5 dB / 1.62 MHz / 74.7°	-60.2 / -55.5 dB
MP-DDA, MOM	100 p	0.5000	1.02 mV	96.5 dB / 2.03 MHz / 66.2°	-60.0 / -59.8 dB

variant	load	gain	nonlin	T0 / fc / PM	THD 10 k / 100 kHz
MP-DDA, hv-PMOS accumulation cap 14 × 14 μm	100 p	0.5000	1.02 mV	96.7 dB / 2.80 MHz / 53.4°	−60.0 / −59.9 dB

Same current (427 μA) and tail budget ($8 \times 10 \mu\text{A} = \text{baseline } 4 \times 20 \mu\text{A}$). 9 dB better THD, 5× lower DC nonlinearity. The MOS cap (196 μm² vs 961 μm² MOM) does not degrade THD despite a 2× C–V variation over 0.15...1.25 V ($0.37 \rightarrow 0.75 \text{ pF}$); it is just smaller than 1 pF on average, so it needs resizing for the PM.

14:48 — MOS capacitor sizing, corners, mismatch

MOS accumulation cap size (MP-DDA, f_c / PM): 18 × 18 μm matches the 31 × 31 μm MOM pair almost exactly (100 pF: 2.00 MHz / 67.0° vs 2.03 MHz / 66.2°; 300 pF: 46.3° vs 45.5°; 1 nF: 26.9° vs 26.4°). Area per side 324 μm² instead of 961 μm² (3.0×), no metal used.

Corners (dc gain fitted over $|v_d| \leq 0.2 \text{ V}$, 1 kΩ || 100 pF; nonlin = max deviation over ±1 V):

corner	baseline A gain / nonlin	MP-DDA gain / nonlin
tt 27	0.5005 / 5.52 mV	0.4998 / 1.04 mV
ss 125 (res wcs)	0.5119 / 8.07	0.4988 / 1.56
ff −40 (res bcs)	0.4968 / 3.30	0.4998 / 0.66
sf / fs 27	0.4998 / 4.11, 0.5013 / 8.07	0.4998 / 0.82, 0.4997 / 1.37
ss 27 / ff 27	0.5022 / 12.69, 0.4990 / 3.30	0.4998 / 1.91, 0.4997 / 0.69
tt res wcs / bcs	0.4895 / 7.09, 0.5148 / 5.07	0.4997 / 1.48, 0.4998 / 0.66
tt 125 / tt −40	0.5249 / 5.18, 0.4870 / 7.15	0.4983 / 1.10, 0.4998 / 1.53

Systematic gain spread $-2.6 \dots +5.0 \%$ → $-0.34 \dots 0 \%$. Loop gain / PM unchanged. (Corrected 15:25: this line first said “±3.8 %”, which was half the spread, not its range.)

Mismatch MC (30 seeds, `mos_tt_mismatch` + `res_typ_mismatch`, `mm_ok=1` on all MOS and rhigh): baseline offset σ 14.4 mV, MP-DDA σ 14.5 mV — the front-end change does not touch the offset. Budget by device group (mismatch on one group at a time, 20 seeds): folding sinks 11.5 mV, tails 4.2, mirror 2.3, pairs 1.7, everything else < 0.4. Same mechanism Sarpeshkar describes (§3.3.4: current mismatch / transconductance; low Gm/I front ends are offset-prone, and the mirrors, not the input devices, matter most).

14:53 — d2s_mp2: same topology, resized for current and offset

- units at $2 \times 5 \mu\text{A}$ (long-L tails 5/6), $R = 150 \text{ k}\Omega$ (rhigh $0.5 \times 50.6 \mu\text{m}$), $I_{tR} = 0.75 \text{ V}$
- folding sinks 30 μA, 36/6 (lower gm/I), mirror 24/4 at 10 μA
- bias fixture `d2s_bias_lp`: reference currents $5 + 5 + 2 + 2 + 5 + 5 \mu\text{A}$ instead of $20 + 50 + 5 + 5 + 5 + 5$
- headroom limit found on the way: $a \approx 2.6 \text{ V}$ and $b \approx 0.7 \text{ V}$ are the output-device gates, so the mirror and the sinks each get only $\sim 0.35 \text{ V}$; their overdrive can't be raised for matching, only their area.
- all devices $\geq 70 \text{ mV}$ from saturation edge at $v_d = 0$ (`oppoint.py`)

	baseline A	d2s_mp2, MOS 16 × 16
I _{DD} (incl. bias fixture)	427 μA	297 μA
I _Q P/N	240 μA	213 μA
compensation area	$2 \times 961 \mu\text{m}^2$ MOM (M1–M4)	$2 \times 256 \mu\text{m}^2$ hv-PMOS
T0 1 k 100 p / 100 p / 50 Ω	66.5 / 96.5 / 40.7 dB	69.7 / 99.6 / 43.9 dB
fc / PM, 100 p	2.02 MHz / 66°	1.76 MHz / 63°

	baseline A	d2s_mp2, MOS 16×16
THD 10 kHz (1 k $\parallel$ 100 p)	−51.1 dB	−58.7 dB
THD 100 kHz (1 k $\parallel$ 100 p)	−50.2 dB	−51.4 dB
nonlin (± 1 V vd)	5.52 mV	1.24 mV
offset σ (mismatch, 30)	14.4 mV	5.3 mV
gain σ (mismatch, 30)	1.6 %	1.1 % (0.84 % from rhigh unit mismatch)

Gain σ is now set by rhigh area ($0.5 \times 50.6 \mu\text{m}$ per unit): $4\times$ the area halves it.

14:57 — the right figure of merit for a DDA unit (sim/run_improvements.py units)

The first unit test swept both inputs around a common CM. In the MP-DDA one input of every unit sits at vref, so the relevant comparison is $f_A(x) = I(\text{vref} + x, \text{vref})$ against $f_B(x) = I(\text{vref}, \text{vref} - x)$: the same differential voltage, but the pair's sources sit x lower in B. Solving $f_A(u) + f_B(u) = 2 f_A(v)$ from the unit curves alone predicts the whole driver's static error; for unit_r it gives $-1.04 / -1.02$ mV at $u = \mp 0.5$ V, which is the full-circuit 1.04 mV.

unit	own compression at 0.5 V	MP-DDA error at $u = -0.5 / +0.5$ V
unit_r (split tails + rhigh)	0.5 %	−1.0 / −1.0 mV
unit_t (triode PMOS)	20 %	−191 / −72 mV
unit_w (well input)	7 %	+29 / +38 mV
unit_q (undegenerated 2/6, square law, no R)	22 %	+75 / +200 mV

So the criterion for an ELIN unit here is not its linearity but that its output depends on $g_p - g_n$ only. The split-tail pair meets it because each device is a constant-current source follower (Minch thesis ch. 7: a follower is linear whatever the device law) and the element between the sources sees exactly $g_p - g_n$. Any element whose value depends on the absolute source voltage (triode R, well κ , tail/CLM of a single-tail pair) fails.

14:56 — cascoding

(a) output devices = ESD clamp frames. A series cascode would put its drain, not the clamp's, on the pad; a stacked-clamp frame would be a new ESD structure to qualify. So no output cascode. What remains is cascoding the internal high-impedance nodes. Lengthening the folded and mirror cascodes ($L_1 \rightarrow 3 \mu\text{m}$, W scaled, same overdrive) in d2s_mp2:

cascade L	T0 1 k $\parallel$ 100 p	T0 100 p	T0 50 Ω	PM 100 p
1 μm (d2s_mp2)	69.7 dB	99.6 dB	43.9 dB	63.4°
2 μm	75.4	103.1	49.6	62.8°
3 μm (d2s_mp2c3)	78.9	104.3	53.1	62.0°

+9 dB for $\sim 320 \mu\text{m}^2$ of gate area. Regulated (gain-boosted) cascodes were not tried: node a sits at ~ 2.6 V and b at ~ 0.7 V (the output gates), which leaves ~ 0.35 V per device for the mirror, the sinks and their cascodes, so no headroom for an auxiliary amplifier's V_{GS} .

(b) separate output devices in the slot. Then they can be cascoded. Test case d2s_lc2: load-compensated (topology B, no compensation capacitor at all), MP-DDA front end, half-frame output devices, cascode gates simply at vref (output range $\sim 1.0 \dots 2.4$ V):

	T0	5 p	20 p	100 p	1 n	1 k 100 p	I_DD
no cascode	37.2 dB	16°	47°	78°	89°	no loop gain	174 μ A
cascoded	74.0 dB	unstable	39°	77°	89°	no loop gain	169 μ A

Cascoding adds 37 dB; a resistive load still collapses a single-stage loop, and the minimum load capacitance stays (~ 30 – 40 pF). In the slot, the pad goes through IOPadAnalog, whose **padres** is behind $587\ \Omega$ (SecondaryProtection). Sensing at **padres** (near side) makes that resistor an isolation resistor: PM at $20\text{ pF } 39^\circ \rightarrow 62^\circ$, $100\text{ pF } \rightarrow 104^\circ$; the pad then sees a $587\ \Omega \cdot \text{C.L.}$ low-pass outside the loop (2.7 MHz at 100 pF). Sensing at **pad** keeps the PM of the no-resistor case. $50\ \Omega$ through $587\ \Omega$ is not drivable at all. Open point: the cascoded version has 7.8 mV systematic offset (diode replicas at a different V_{DS} than the cascoded output devices); cascoding the diode replicas would fix it.

14:58 — noise (1 k || 100 p, output-referred)

	1 kHz	100 kHz	10 Hz–10 MHz
baseline A	1599 nV/ $\sqrt{\text{Hz}}$	200 nV/ $\sqrt{\text{Hz}}$	278 $\mu\text{V rms}$
MP-DDA (baseline sizing)	1592	201	279
d2s_mp2c3	676	152	231

14:59 — d2s_mp2c3 (= d2s_mp2 with $L = 3\ \mu\text{m}$ cascodes, MOS caps $16 \times 16\ \mu\text{m}$): full table

Renamed `d2s_mpd` (`sim/d2s_mpd.spice`); every number re-run in `sim/results_improvements.txt`. 296 μA (baseline 427), T0 78.9 dB at $1\text{ k} || 100\text{ p}$ (66.5), 53.1 dB at $50\ \Omega$ (40.7), 104 dB capacitive (96.5), PM 62° at 100 pF , 42° at 300 pF . Gain 0.4992...0.5000 over 11 corners, $\text{nonlin} \leq 3.5\text{ mV}$ (ss 27°C). Offset $\sigma 5.3\text{ mV}$, gain $\sigma 1.1\%$ (rhig area). THD 10 kHz -58.5 dB ; 100 kHz -51.3 dB ($1\text{ k} || 100\text{ p}$; lower fc than the MOM-MP version, 1.37 vs 1.62 MHz).

15:05–15:20 — consolidation

Intermediate netlists and scripts folded into one runner, `sim/run_improvements.py`, which reuses `../run_d2s.py` unchanged and reproduces every number of this log and of `proposed_improvements.md` in one run (3 min 49 s, `sim/results_improvements.txt`). The re-run matches the numbers above. Files in `sim/`:

file	content
<code>units.spice</code>	DDA units: <code>unit_r</code> (rhig), <code>unit_r2</code> (final, $2 \times 5\ \mu\text{A}$), <code>unit_t</code> (triode MOS), <code>unit_w</code> (well input), <code>unit_q</code> (undegenerated)
<code>d2s_mp.spice</code>	MP-DDA in the baseline's sizing; compensation through the <code>ccomp</code> wrapper
<code>ccomp.spice</code>	<code>ccomp_mom</code> (as <code>d2s_miller</code>), <code>ccomp_mos</code> (hv PMOS accumulation caps)
<code>d2s_mpd.spice</code>	the proposal (frames kept as output devices)
<code>d2s_bias_lp.spice</code>	bias fixture with $26\ \mu\text{A}$ of reference current instead of $90\ \mu\text{A}$
<code>d2s_lc2.spice</code>	case (b): in-slot output devices, load-compensated, cascoded (and not)
<code>run_improvements.py</code> , <code>results_improvements.txt</code>	runner and its output

15:18 — transfer check

Checksums of the committed files against the container copies: `proposed_improvements.md`, `sim/results_improvements.txt` and `sim/run_improvements.py` did not match after the first commits, although the tool reported "written". The device still held earlier versions of them, 20530 / 16367 / 13939 bytes instead of 20571 / 18250 / 16300. Re-committing from the same staged paths, with force, changed nothing. Committing copies from a *new* staged directory (`outputs/xfer2/`) worked. All files now match by sha256. This looks like the silent no-op described in the project CLAUDE.md, §6. Workaround: stage every new version of a file under a new staged path.

15:20 — time and tokens used by this task

Wall clock: 14:08 → ~15:20, about 72 minutes, interrupted twice by questions. Of that, the compute (not the model) took about 3 min for the OpenVAF build and model compile, about 6 min for OCR on your computer, and about 10 min of ngspice in total, including the 3 min 49 s full re-run.

Tokens, from this session's own transcript (`~/.claude/projects/-home-claude/<session>.jsonl`, per API response, as of 15:19):

phase	wall clock	API calls	output tokens	cache writes	cache reads
orientation: existing note, netlists, CACE results, CLAUDE.md	14:08–14:12	14	20,138	106,368	1,958,053
simulation environment + literature reading/OCR, interleaved	14:12–14:27	49	23,126	122,848	12,499,286
baseline reproduced; Caltech-thesis check	14:27–14:31	6	2,887	13,684	1,960,595
design reasoning + simulations	14:31–15:00	40	125,771	157,464	17,134,286
accounting, log timestamp correction	15:00–15:02	5	10,264	17,604	2,485,393
consolidated runner, re-run, notes, proposal, transfer checks	15:02–15:19	37	60,208	98,924	21,839,596
total, main thread		151	242,394	517,090	56,877,712
subagent (Caltech PDF check)	14:30–14:31	5	1,463	94,882	368,538

How to read it:

- **Output tokens** (~244 k): what the model wrote, including its reasoning. More than half of it falls in the design phase.
- **Cache reads** (~57 M): each API call re-reads the whole conversation so far from the prompt cache. The context grew from ~118 k tokens at the start (system prompt, tools, memory) to ~608 k at the end. The growth came mostly from tool output: paper text, OCR, netlists and simulation listings.
- **Cache writes** (~0.5 M): new context added to the cache.
- **Uncached input**: 302 tokens in all.

How cache reads, cache writes and output count against a specific plan's usage limits is not something this log can state (not checked here); the plans' usage limits are documented at support.claude.com. What this kind of task pushes is the context size and the number of calls made over it.

What drove the size, for planning similar tasks:

- Reading papers as text into the conversation: Edwards, Minch and Hasler excerpts and the Sarpeshkar OCR, roughly 100–150 k tokens of context.

- Long simulation listings.
- A long reasoning phase before the first simulation.

A second session that starts from `proposed_improvements.md` and `log.md` instead of the papers would begin with far less context.

15:55 — IEEEtran version

`proposed_improvements.tex` is `proposed_improvements.md` in IEEEtran journal style: the same content and numbers, with equations and a references list taken from `literature_notes.md`. `proposed_improvements.pdf` (5 pages) was built in the cloud with `pdflatex`, run twice, and had no errors and no overfull boxes. The only packages it needs besides `IEEEtran.cls` are `amsmath`, `booktabs`, `tabularx`, `array`, `textcomp`, `url` and `cite`; there is no `siunitx`. The sandbox I reach on the linked computer has `pdflatex` but neither `IEEEtran.cls` nor `siunitx`. Building there needs TeX Live's `texlive-publishers`, or a copy of `IEEEtran.cls` next to the file.

16:06 — time and tokens, updated through the LaTeX work

Same method as at 15:20, now including the report, the questions afterwards and the two LaTeX conversions:

phase	wall clock	API calls	output tokens	cache writes	cache reads
orientation	14:08–14:12	14	20,138	106,368	1,958,053
simulation environment + literature/OCR	14:12–14:27	49	23,126	122,848	12,499,286
baseline reproduced; Caltech-thesis check	14:27–14:31	6	2,887	13,684	1,960,595
design reasoning + simulations	14:31–15:00	40	125,771	157,464	17,134,286
accounting, log timestamp correction	15:00–15:02	5	10,264	17,604	2,485,393
runner, re-run, notes, proposal, transfer checks, report	15:02–15:20	41	64,333	102,341	23,278,772
questions about the spreadsheet button	15:20–15:52	1	427	6,984	613,327
IEEEtran LaTeX of the proposal	15:52–16:04	16	32,152	45,403	10,226,268
resource-log question; start of the log's LaTeX	16:04–16:06	2	1,576	7,552	1,339,858
total, main thread		174	280,674	580,248	71,495,838
subagent (Caltech PDF check)	14:30–14:31	5	1,463	94,882	368,538

The context is now ~674 k tokens per call. The LaTeX conversion of this log comes after this table and is not in it.

16:15 — LaTeX version of this log

`log.tex` / `log.pdf` (article class, not IEEEtran, in the style of `sudelbuecher/esd_protection/*.tex`). The body was converted from this file with `pandoc 3.1.3`, and the tables were given widths weighted by their content. Builds with `pdflatex` using only standard packages (`booktabs`, `longtable`, `calc`, `hyperref`, `xurl`, `microtype`, `textcomp`, `geometry`). A snapshot: entries after 16:15 are only in `log.md`.